Figure 4.4 Progression pattern and post-progression cause of death

Analysis cohort: 1786 patients with first radiological progression (HAIC-Sys 733, TACE-Sys 613, Systemic 440); 1186 deaths after progression.
Deaths by cause: intrahepatic HCC 696, extrahepatic HCC 130, systemic HCC 272, non-HCC 80, undetermined 8.
Median post-progression follow-up (reverse KM): 18.7 months. Time zero = date of first progression; main time point 12 months.
Bootstrap: 1000 successful patient-level resamples (all models refitted in each). Engine check vs riskRegression: max diff 7.2e-16.

Contents:

1. Figure 4.4A1-A4 adjusted 12-month CIF matrix; ARD vs population average; cause composition at 6/12/24 months; PMR enrichment
2. Figure 4.4B1-B2 csHR and sHR matrices (reference Sys-TG) + overall Wald; Lunn-McNeil heterogeneity tests
3. Cohort flow, events and numbers at risk; 4. Lunn-McNeil and follow-up tests; 5. Overall Wald tests, PH tests
6. Validation vs original IH-TG matrix (incl. earlier pattern rule); 7. Treatment interaction; 8+. Supplementary figures (sHR, ARD, PMR, unadjusted AJ, 6/24 months, Firth, sensitivity)

A Adjusted 12-month cumulative incidence of death after first progression

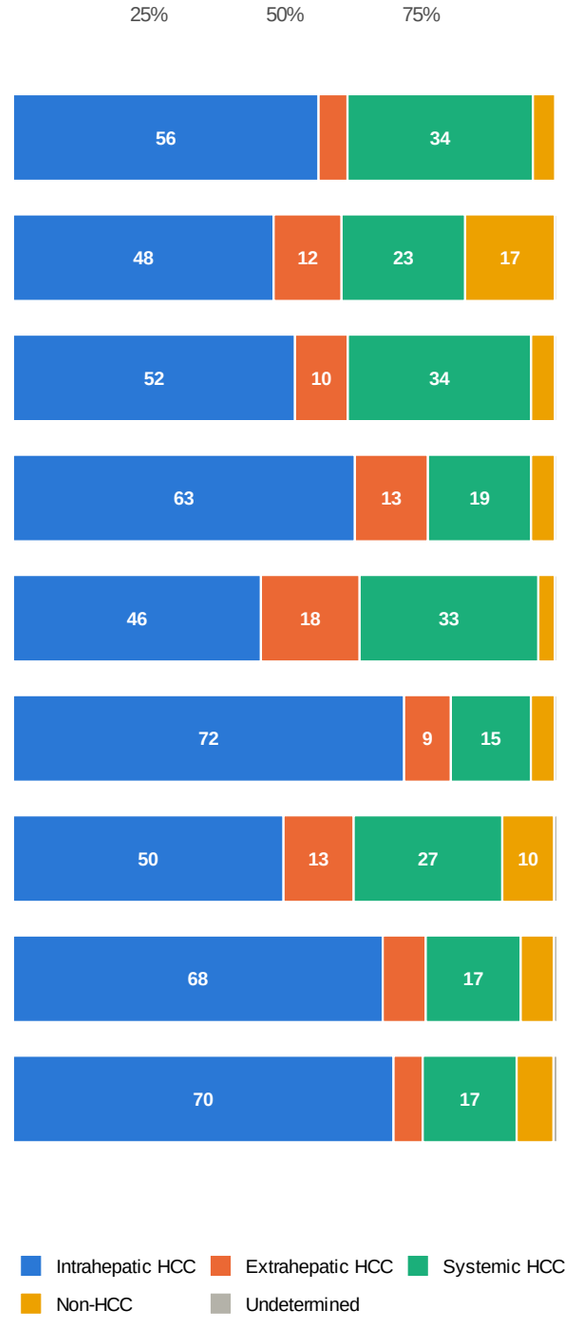
Cause-specific Cox + G-computation (standardised to all 1786 progressors); % (95% bootstrap CI, B = 1000); n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	77.5 (65.1–88.6) n=36	43.5 (29.7–57.6) n=18	4.1 (0.0–11.5) n=3	26.4 (14.7–40.5) n=14	3.2 (0.0–11.2) n=1
EH-TG+NL	75.4 (63.1–87.5) n=21	36.1 (20.3–51.3) n=10	9.4 (0.0–20.5) n=2	17.1 (6.5–31.3) n=6	12.5 (0.0–29.7) n=3
Sys-TG+NL	75.2 (69.6–81.3) n=154	39.0 (33.3–45.6) n=88	7.3 (3.9–11.7) n=13	25.3 (19.5–31.0) n=47	3.3 (0.9–6.4) n=6
EH-TG	74.9 (67.7–81.8) n=72	47.0 (37.6–55.9) n=42	10.1 (5.2–15.8) n=11	14.2 (8.2–21.1) n=15	3.3 (0.6–8.0) n=3
Sys-NL	74.0 (67.3–81.3) n=63	33.7 (24.9–42.1) n=26	13.4 (6.9–22.6) n=13	24.3 (15.6–33.3) n=22	2.3 (0.0–6.2) n=2
IH-TG+NL	71.7 (64.7–78.8) n=74	51.5 (43.8–59.6) n=55	6.2 (2.1–11.1) n=6	10.5 (5.3–16.0) n=10	3.2 (0.7–6.5) n=3
EH-NL	64.9 (60.5–69.5) n=159	32.2 (27.4–37.2) n=76	8.4 (5.5–11.4) n=18	17.7 (13.4–22.5) n=48	6.2 (3.9–9.2) n=15
IH-TG	62.1 (58.6–66.2) n=269	42.1 (38.0–46.5) n=193	4.9 (3.0–7.0) n=19	10.8 (8.1–13.9) n=38	3.8 (2.2–5.6) n=16
IH-NL	58.0 (53.5–62.6) n=161	40.5 (35.3–45.8) n=109	3.1 (1.6–4.7) n=8	10.0 (7.1–13.4) n=29	3.9 (2.3–6.0) n=15

12-month CIF (%)

0 25 50 75

Cause composition at 12 months

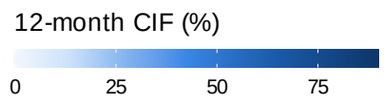


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. Rows ordered by adjusted all-cause CIF. Cause-specific CIFs plus undetermined deaths sum to all-cause. Composition labels show % of deaths by 12 months (segments <8% unlabelled).

A1 Adjusted 12-month cumulative incidence of death after first progression

Cause-specific Cox + G-computation (standardised to all 1786 progressors); % (95% bootstrap CI, B = 1000); n = events within 12 months

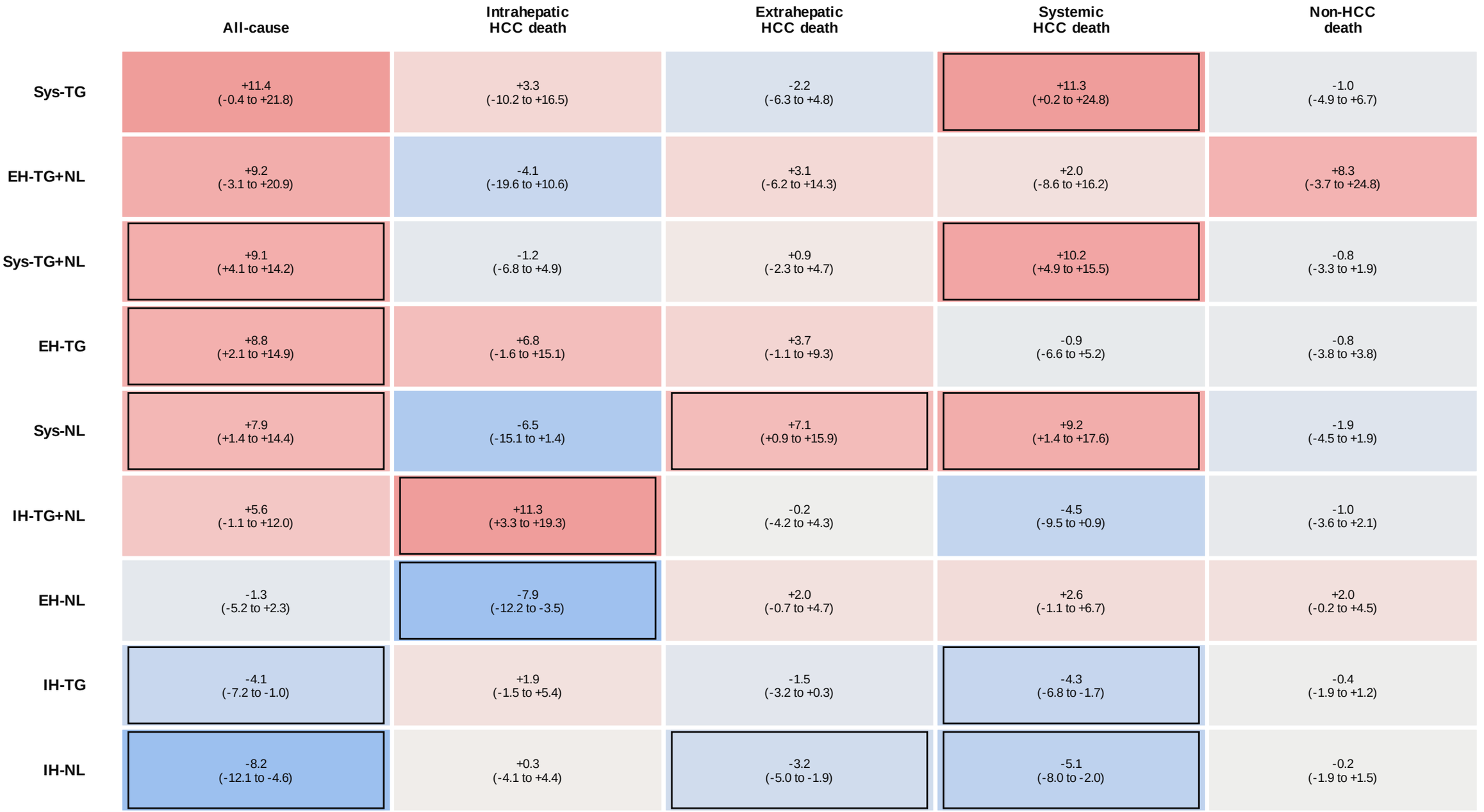
	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	77.5 (65.1–88.6) n=36	43.5 (29.7–57.6) n=18	4.1 (0.0–11.5) n=3	26.4 (14.7–40.5) n=14	3.2 (0.0–11.2) n=1
EH-TG+NL	75.4 (63.1–87.5) n=21	36.1 (20.3–51.3) n=10	9.4 (0.0–20.5) n=2	17.1 (6.5–31.3) n=6	12.5 (0.0–29.7) n=3
Sys-TG+NL	75.2 (69.6–81.3) n=154	39.0 (33.3–45.6) n=88	7.3 (3.9–11.7) n=13	25.3 (19.5–31.0) n=47	3.3 (0.9–6.4) n=6
EH-TG	74.9 (67.7–81.8) n=72	47.0 (37.6–55.9) n=42	10.1 (5.2–15.8) n=11	14.2 (8.2–21.1) n=15	3.3 (0.6–8.0) n=3
Sys-NL	74.0 (67.3–81.3) n=63	33.7 (24.9–42.1) n=26	13.4 (6.9–22.6) n=13	24.3 (15.6–33.3) n=22	2.3 (0.0–6.2) n=2
IH-TG+NL	71.7 (64.7–78.8) n=74	51.5 (43.8–59.6) n=55	6.2 (2.1–11.1) n=6	10.5 (5.3–16.0) n=10	3.2 (0.7–6.5) n=3
EH-NL	64.9 (60.5–69.5) n=159	32.2 (27.4–37.2) n=76	8.4 (5.5–11.4) n=18	17.7 (13.4–22.5) n=48	6.2 (3.9–9.2) n=15
IH-TG	62.1 (58.6–66.2) n=269	42.1 (38.0–46.5) n=193	4.9 (3.0–7.0) n=19	10.8 (8.1–13.9) n=38	3.8 (2.2–5.6) n=16
IH-NL	58.0 (53.5–62.6) n=161	40.5 (35.3–45.8) n=109	3.1 (1.6–4.7) n=8	10.0 (7.1–13.4) n=29	3.9 (2.3–6.0) n=15



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. Rows ordered by adjusted all-cause CIF; 0 events: no event by the time point (no model estimate shown).

A2 Absolute risk difference at 12 months vs population-standardised average

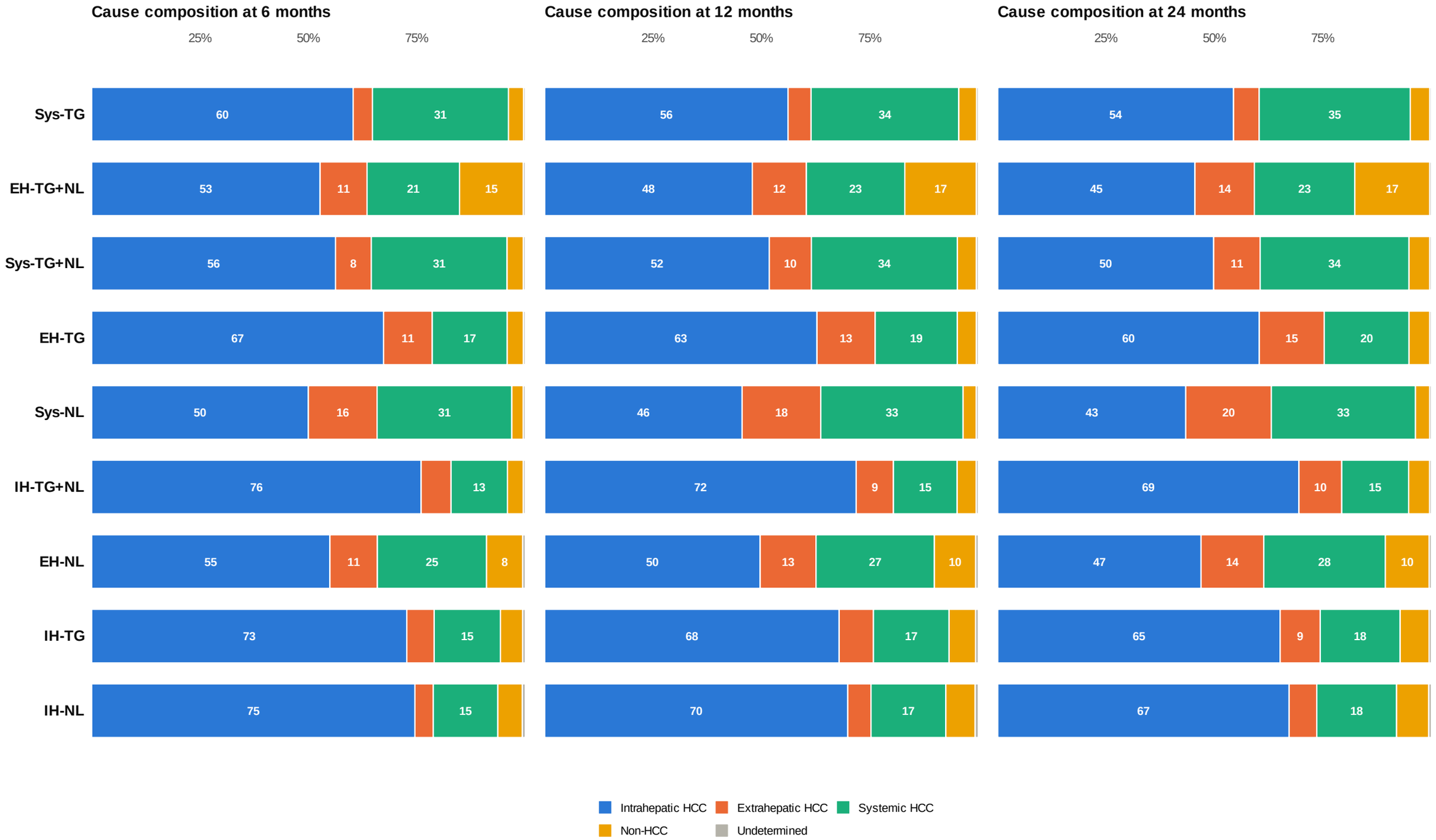
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

A3 Cause-of-death composition at fixed time points after first progression

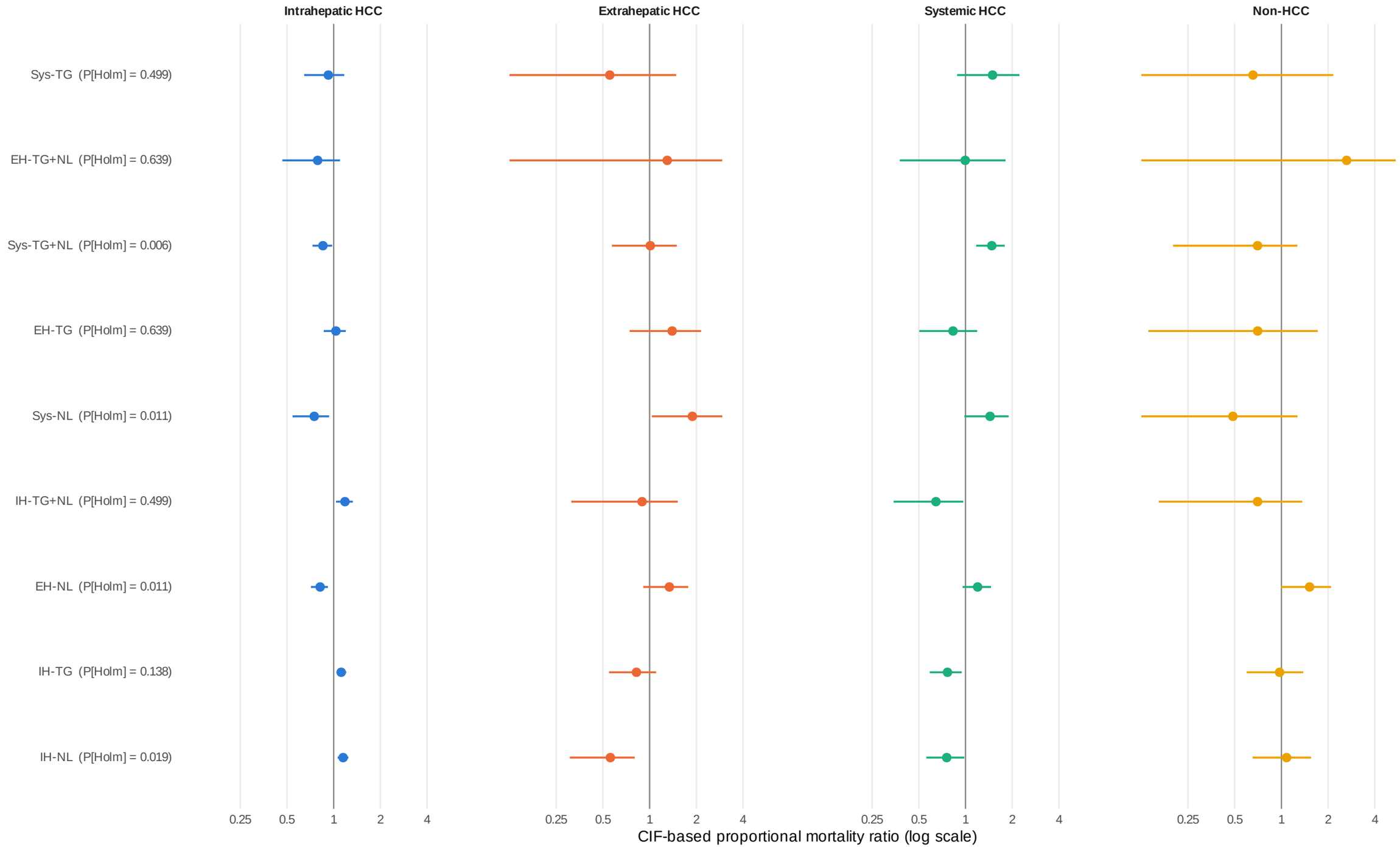
Q = cause-specific standardised CIF / all-cause standardised CIF (sums to 100% within each pattern); labels = % (segments <8% unlabelled)



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. Rows ordered by adjusted 12-month all-cause CIF.

A4 Cause-of-death enrichment: CIF-based proportional mortality ratio (PMR) at 12 months

PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).

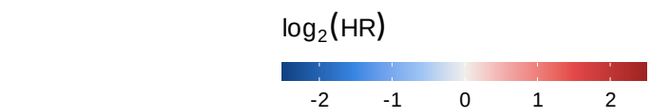


PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

B1 Cause-specific hazard ratios by first-progression pattern (reference: Sys-TG)

All-cause: Cox HR; cause-specific: csHR (cause-specific Cox). Lunn–McNeil heterogeneity across IH/EH/Systemic HCC death: pattern × cause $\chi^2(16) = 60.2$, $P < 0.001$; site × cause $\chi^2(4) = 48.8$, $P < 0.001$; form × cause $\chi^2(4) = 7.7$, $P = 0.104$

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.58 (0.41–0.82)**	0.71 (0.44–1.15)	0.88 (0.26–3.04)	0.28 (0.16–0.51)***	NE (sparse reference)
IH-NL	0.50 (0.35–0.70)***	0.64 (0.39–1.05)	0.52 (0.15–1.88)	0.24 (0.13–0.45)***	NE (sparse reference)
IH-TG+NL	0.79 (0.53–1.16)	1.07 (0.63–1.80)	1.32 (0.33–5.30)	0.34 (0.16–0.72)**	NE (sparse reference)
EH-TG	0.91 (0.62–1.34)	1.01 (0.59–1.75)	2.33 (0.65–8.35)	0.49 (0.24–0.98)*	NE (sparse reference)
EH-NL	0.62 (0.44–0.89)**	0.54 (0.33–0.89)*	1.58 (0.47–5.34)	0.47 (0.27–0.84)*	NE (sparse reference)
EH-TG+NL	0.85 (0.50–1.44)	0.73 (0.34–1.59)	2.15 (0.41–11.31)	0.56 (0.22–1.43)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.84 (0.56–1.26)	0.66 (0.37–1.20)	3.07 (0.84–11.18)	0.80 (0.42–1.55)	NE (sparse reference)
Sys-TG+NL	0.88 (0.62–1.26)	0.82 (0.50–1.36)	1.69 (0.48–5.90)	0.88 (0.50–1.56)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.405**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. NE: <3 events in the cell or in the reference cell. * P<0.05, ** P<0.01, *** P<0.001 (unadjusted; BH-FDR q values in Table S-HR). Overall Wald test of the 8 pattern terms does not depend on the reference.

B1 Subdistribution hazard ratios (Fine–Gray) by progression pattern (reference: Sys-TG)

Fine–Gray via survival::finegray with stratum-specific censoring weights; robust SE

	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	1.05 (0.63–1.77)	1.07 (0.30–3.74)	0.38 (0.21–0.70)**	NE (sparse reference)
IH-NL	0.97 (0.57–1.65)	0.71 (0.19–2.65)	0.33 (0.17–0.62)***	NE (sparse reference)
IH-TG+NL	1.45 (0.82–2.58)	1.29 (0.32–5.19)	0.37 (0.17–0.79)*	NE (sparse reference)
EH-TG	1.06 (0.59–1.93)	2.37 (0.64–8.77)	0.45 (0.23–0.92)*	NE (sparse reference)
EH-NL	0.69 (0.40–1.19)	1.98 (0.58–6.83)	0.63 (0.34–1.14)	NE (sparse reference)
EH-TG+NL	0.79 (0.35–1.78)	1.84 (0.36–9.31)	0.64 (0.25–1.60)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.72 (0.39–1.33)	3.10 (0.81–11.89)	0.88 (0.45–1.72)	NE (sparse reference)
Sys-TG+NL	0.96 (0.55–1.67)	1.48 (0.41–5.35)	0.90 (0.49–1.66)	NE (sparse reference)
Overall Wald P (df=8)	**P = 0.004**	**P = 0.003**	**P < 0.001**	**P = 0.265**

$\log_2(\text{sHR})$

-2 -1 0 1 2

Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. NE: <3 events in the cell or in the reference cell. Overall Wald test (df = 8, robust variance) is reference-independent.

B2 Lunn–McNeil test of whether pattern effects differ across intrahepatic / extrahepatic / systemic HCC death

Overall likelihood-ratio tests (reference-independent; non-HCC and undetermined deaths censored; strata = cause x treatment group; all covariates interact with cause)

Test	LR chi-square	df	P
Pattern x cause (9 patterns)	60.17	16	<0.001
Site x cause (IH / EH / Sys)	48.80	4	<0.001
Form x cause (TG / NL / TG+NL)	7.69	4	0.104

Follow-up (weighted effect coding): csHR of each pattern vs the pattern-weighted average, per cause; Wald test that the effect is equal across the three causes

Pattern	IH death	EH death	Systemic death	Wald chi2 (df=2)	P	P (Holm)
IH-TG	0.99 (0.88-1.12)	0.78 (0.55-1.10)	0.70 (0.56-0.89)	7.19	0.028	0.138
IH-NL	0.90 (0.77-1.05)	0.46 (0.30-0.72)	0.61 (0.45-0.81)	11.50	0.003	0.019
IH-TG+NL	1.49 (1.16-1.91)	1.17 (0.57-2.42)	0.85 (0.50-1.45)	3.62	0.164	0.499
EH-TG	1.41 (1.05-1.90)	2.07 (1.22-3.50)	1.22 (0.77-1.94)	2.28	0.319	0.639
EH-NL	0.75 (0.62-0.91)	1.40 (0.98-1.99)	1.18 (0.91-1.53)	13.21	0.001	0.011
EH-TG+NL	1.02 (0.55-1.91)	1.90 (0.59-6.16)	1.41 (0.65-3.06)	0.98	0.613	0.639
Sys-TG	1.39 (0.89-2.19)	0.88 (0.28-2.78)	2.50 (1.50-4.16)	4.16	0.125	0.499
Sys-NL	0.93 (0.64-1.33)	2.72 (1.55-4.76)	2.00 (1.32-3.04)	12.89	0.002	0.011
Sys-TG+NL	1.14 (0.94-1.40)	1.49 (0.93-2.39)	2.20 (1.68-2.89)	14.52	<0.001	0.006

Cohort flow

step	n
Total patients	4204
PFS event (progression or death)	2239
excluded: death without progression	441
First radiological progression	1798
excluded: PPOS missing (PFS > OS, data error)	1
excluded: flagged for deletion (non-HCC histology / other cancer / external treatment / long LTFU)	11
Analysis cohort	1786

Progression-pattern classification by component combination (analysis cohort)

(1) IH-TG only IHG; (2) IH-NL only NIH; (3) IH-TG+NL IHG+NIH; (4) EH-TG only EHG; (5) EH-NL only NEH; (6) EH-TG+NL EHG+NEH; (7) Sys-TG only STG; (8) Sys-NL NIH+NEH; (9) Sys-TG+NL all other combinations

Components (TTU->IHG, nMVI->NIH)	Pattern	Earlier rule	n	
IHG	IH-TG	IH-TG	310	
TTU	IH-TG	IH-TG	89	
TTU + IHG	IH-TG	IH-TG	77	
NIH	IH-NL	IH-NL	356	
nMVI	IH-NL	IH-NL	16	
NIH + IHG	IH-TG+NL	IH-TG+NL	51	
TTU + NIH	IH-TG+NL	IH-TG+NL	29	
nMVI + IHG	IH-TG+NL	IH-TG+NL	18	
TTU + NIH + IHG	IH-TG+NL	IH-TG+NL	16	
nMVI + NIH + IHG	IH-TG+NL	IH-TG+NL	2	
EHG	EH-TG	EH-TG	125	
NEH	EH-NL	EH-NL	298	
NEH + EHG	EH-TG+NL	EH-TG+NL	34	
STG	Sys-TG	Sys-TG	52	
NIH + NEH	Sys-NL	Sys-NL	87	
nMVI + NEH	Sys-NL	Sys-NL	5	
nMVI + NIH + NEH	Sys-NL	Sys-NL	2	
IHG + NEH	Sys-TG+NL	Sys-TG+NL	59	
TTU + IHG + NEH	Sys-TG+NL	Sys-TG+NL	22	
TTU + NEH	Sys-TG+NL	Sys-TG+NL	20	
TTU + STG	Sys-TG+NL	Sys-TG	14	<- reclassified
TTU + NIH + NEH	Sys-TG+NL	Sys-TG+NL	11	
STG + NEH	Sys-TG+NL	Sys-TG+NL	10	
TTU + STG + NEH	Sys-TG+NL	Sys-TG+NL	9	

Progression-pattern classification by component combination (analysis cohort)

(1) IH-TG only IHG; (2) IH-NL only NIH; (3) IH-TG+NL IHG+NIH; (4) EH-TG only EHG; (5) EH-NL only NEH; (6) EH-TG+NL EHG+NEH; (7) Sys-TG only STG; (8) Sys-NL NIH+NEH; (9) Sys-TG+NL all other combinations

Components (TTU->IHG, nMVI->NIH)	Pattern	Earlier rule	n	
NIH + EHG	Sys-TG+NL	Sys-TG+NL	9	
TTU + EHG	Sys-TG+NL	Sys-TG	8	<- reclassified
TTU + NEH + EHG	Sys-TG+NL	Sys-TG+NL	8	
NIH + NEH + EHG	Sys-TG+NL	Sys-TG+NL	7	
TTU + NIH + IHG + NEH	Sys-TG+NL	Sys-TG+NL	6	
STG + NIH	Sys-TG+NL	Sys-TG+NL	5	
NIH + IHG + NEH	Sys-TG+NL	Sys-TG+NL	5	
TTU + NIH + EHG	Sys-TG+NL	Sys-TG+NL	4	
nMVI + NIH + NEH + EHG	Sys-TG+NL	Sys-TG+NL	3	
nMVI + NEH + EHG	Sys-TG+NL	Sys-TG+NL	3	
nMVI + EHG	Sys-TG+NL	Sys-TG+NL	3	
nMVI + STG	Sys-TG+NL	Sys-TG+NL	2	
nMVI + STG + NEH	Sys-TG+NL	Sys-TG+NL	2	
STG + NIH + NEH	Sys-TG+NL	Sys-TG+NL	2	
nMVI + STG + NIH	Sys-TG+NL	Sys-TG+NL	1	
nMVI + STG + NIH + NEH	Sys-TG+NL	Sys-TG+NL	1	
nMVI + NIH + IHG + NEH	Sys-TG+NL	Sys-TG+NL	1	
nMVI + NIH + EHG	Sys-TG+NL	Sys-TG+NL	1	
nMVI + IHG + NEH	Sys-TG+NL	Sys-TG+NL	1	
TTU + STG + NIH	Sys-TG+NL	Sys-TG+NL	1	
TTU + NIH + NEH + EHG	Sys-TG+NL	Sys-TG+NL	1	

Deaths after first progression by pattern and cause (whole follow-up) and numbers at risk

IH/EH/Sys = intrahepatic / extrahepatic / systemic HCC-related death; Non-HCC = liver-related + other + treatment-related; Undetermined = unclassified HCC-related + unknown + missing cause

Pattern	N	IH death	EH death	Sys death	Non-HCC	Undetermined	All deaths	At risk 6 mo	At risk 12 mo	At risk 24 mo
IH-TG	476	221	24	49	21	4	319	246	101	35
IH-NL	372	134	15	34	19	1	203	234	134	58
IH-TG+NL	116	63	7	13	4	0	87	55	22	7
EH-TG	125	43	16	19	4	1	83	50	30	11
EH-NL	298	89	31	56	20	2	198	163	75	24
EH-TG+NL	34	10	3	7	3	0	23	13	7	1
Sys-TG	52	19	3	16	1	0	39	10	6	3
Sys-NL	94	28	14	23	2	0	67	34	12	6
Sys-TG+NL	219	89	17	55	6	0	167	57	31	10

Lunn-McNeil cross-cause heterogeneity tests (likelihood ratio)

Expanded data (one row per patient per HCC-related cause); non-HCC and undetermined deaths censored; strata = cause x treatment group; covariates interact with cause

test	chisq	df	p
pattern x cause (9 patterns; IH / EH / Systemic HCC death)	60.17	16	<0.001
site x cause (IH / EH / Sys site), pattern main effect kept	48.80	4	<0.001
form x cause (TG / NL / TG+NL), pattern main effect kept	7.69	4	0.104

Follow-up: pattern effect relative to weighted average pattern (weighted effect coding), per cause

Wald test (df = 2) that the effect-coded csHR is equal across IH/EH/Systemic HCC death; Holm-adjusted across 9 patterns. Descriptive counterpart: PMR (supplementary).

Pattern	IH effect-coded HR	EH effect-coded HR	Sys effect-coded HR	Wald chi2 (df=2)	P	P Holm
IH-TG	0.99 (0.88-1.12)	0.78 (0.55-1.10)	0.70 (0.56-0.89)	7.19	0.028	0.138
IH-NL	0.90 (0.77-1.05)	0.46 (0.30-0.72)	0.61 (0.45-0.81)	11.50	0.003	0.019
IH-TG+NL	1.49 (1.16-1.91)	1.17 (0.57-2.42)	0.85 (0.50-1.45)	3.62	0.164	0.499
EH-TG	1.41 (1.05-1.90)	2.07 (1.22-3.50)	1.22 (0.77-1.94)	2.28	0.319	0.639
EH-NL	0.75 (0.62-0.91)	1.40 (0.98-1.99)	1.18 (0.91-1.53)	13.21	0.001	0.011
EH-TG+NL	1.02 (0.55-1.91)	1.90 (0.59-6.16)	1.41 (0.65-3.06)	0.98	0.613	0.639
Sys-TG	1.39 (0.89-2.19)	0.88 (0.28-2.78)	2.50 (1.50-4.16)	4.16	0.125	0.499
Sys-NL	0.93 (0.64-1.33)	2.72 (1.55-4.76)	2.00 (1.32-3.04)	12.89	0.002	0.011
Sys-TG+NL	1.14 (0.94-1.40)	1.49 (0.93-2.39)	2.20 (1.68-2.89)	14.52	<0.001	0.006

Overall Wald tests of the 8 pattern terms (independent of reference group)

model	outcome	chisq	df	p
cs	All	55.60	8	<0.001
cs	IH	26.24	8	<0.001
cs	EH	30.85	8	<0.001
cs	Sys	61.72	8	<0.001
cs	NonHCC	8.30	8	0.405
fg	IH	22.32	8	0.004
fg	EH	23.01	8	0.003
fg	Sys	39.63	8	<0.001
fg	NonHCC	10.00	8	0.265
firth	All	57.36	8	<0.001
firth	IH	24.43	8	0.002
firth	EH	32.22	8	<0.001
firth	Sys	59.98	8	<0.001
firth	NonHCC	9.49	8	0.303

Proportional hazards (Schoenfeld residuals; cs-Cox / all-cause Cox)

Where the pattern term violates PH, the HR is a follow-up-averaged effect; the fixed-time CIF/ARD (4.4A) do not rely on PH for pattern and are the primary absolute-risk summaries.

outcome	term	chisq	df	p
All	pattern	43.85	8	<0.001
All	GLOBAL	129.04	26	<0.001
IH	pattern	33.05	8	<0.001
IH	GLOBAL	100.78	26	<0.001
EH	pattern	11.87	8	0.157
EH	GLOBAL	26.16	26	0.454
Sys	pattern	8.82	8	0.357
Sys	GLOBAL	41.49	26	0.028
NonHCC	pattern	4.35	8	0.825
NonHCC	GLOBAL	29.50	26	0.289

Validation against the original IH-TG-referenced matrix

Target values from the original PDF: Sys-TG all-cause HR 1.96 (1.48-2.61), Sys-TG csHR systemic death 4.11, IH-NL all-cause HR 0.76

variant	n	deaths	allcause_SysTG	csHR_Sys_SysTG	allcause_IHNL
ORIGINAL matrix (target)	NA	NA	1.96 (1.48-2.61)	4.11	0.76
Main cohort, adjusted, strata(Group)	1786	1186	1.72 (1.22-2.43)	3.54 (1.97-6.37)	0.85 (0.71-1.02)
Earlier pattern rule (Sys-TG n=74), unadjusted	1786	1186	1.88 (1.42-2.50)	4.66 (2.78-7.80)	0.67 (0.56-0.80)
Earlier pattern rule, baseline covariates without TTP	1786	1186	1.93 (1.45-2.57)	3.79 (2.23-6.42)	0.76 (0.63-0.90)
Main cohort, unadjusted, strata(Group)	1786	1186	1.63 (1.17-2.28)	4.53 (2.56-7.99)	0.67 (0.56-0.80)
Incl. flagged pts, adjusted	1797	1197	1.74 (1.23-2.44)	3.58 (1.99-6.44)	0.87 (0.73-1.05)
Incl. flagged pts, unadjusted	1797	1197	1.64 (1.17-2.28)	4.58 (2.59-8.08)	0.68 (0.57-0.81)
Incl. flagged, unadjusted, PPOS_status events	1797	1291	1.51 (1.08-2.10)	4.58 (2.59-8.08)	0.65 (0.55-0.77)
Main cohort, adjusted, PPOS_status events	1786	1280	1.56 (1.11-2.20)	3.54 (1.97-6.37)	0.81 (0.68-0.97)
Main cohort, baseline covariates without TTP	1786	1186	1.77 (1.26-2.48)	3.70 (2.07-6.64)	0.76 (0.63-0.90)
Incl. flagged pts, baseline covariates without TTP	1797	1197	1.78 (1.27-2.50)	3.74 (2.09-6.70)	0.77 (0.65-0.93)

Sensitivity: interaction with treatment group (justifies pooled analysis)

test	chisq	df	p
site x treatment group (all-cause death after progression)	6.92	4	0.140
pattern x treatment group (all-cause death after progression)	24.86	16	0.072

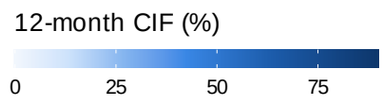
Data linkage checks

check	result
rows (raw / baseline / cod)	4204 / 4204 / 4204
raw.?? == baseline.no == cod.no	TRUE
Group identical in 3 tables	TRUE
baseline.patient_sn == cod.patient_sn	TRUE
raw.Baseline_date == baseline.????????	TRUE
deaths (Death_status_0930 = 1)	2056
deaths with non-missing Cause_of_Mortality	2040
deaths with missing Cause_of_Mortality	16
alive with a recorded Cause_of_Mortality	0
PPOS_status = 1 but Death_status_0930 = 0 (progressors)	94

Supplementary: unadjusted 12-month cumulative incidence (Aalen–Johansen)

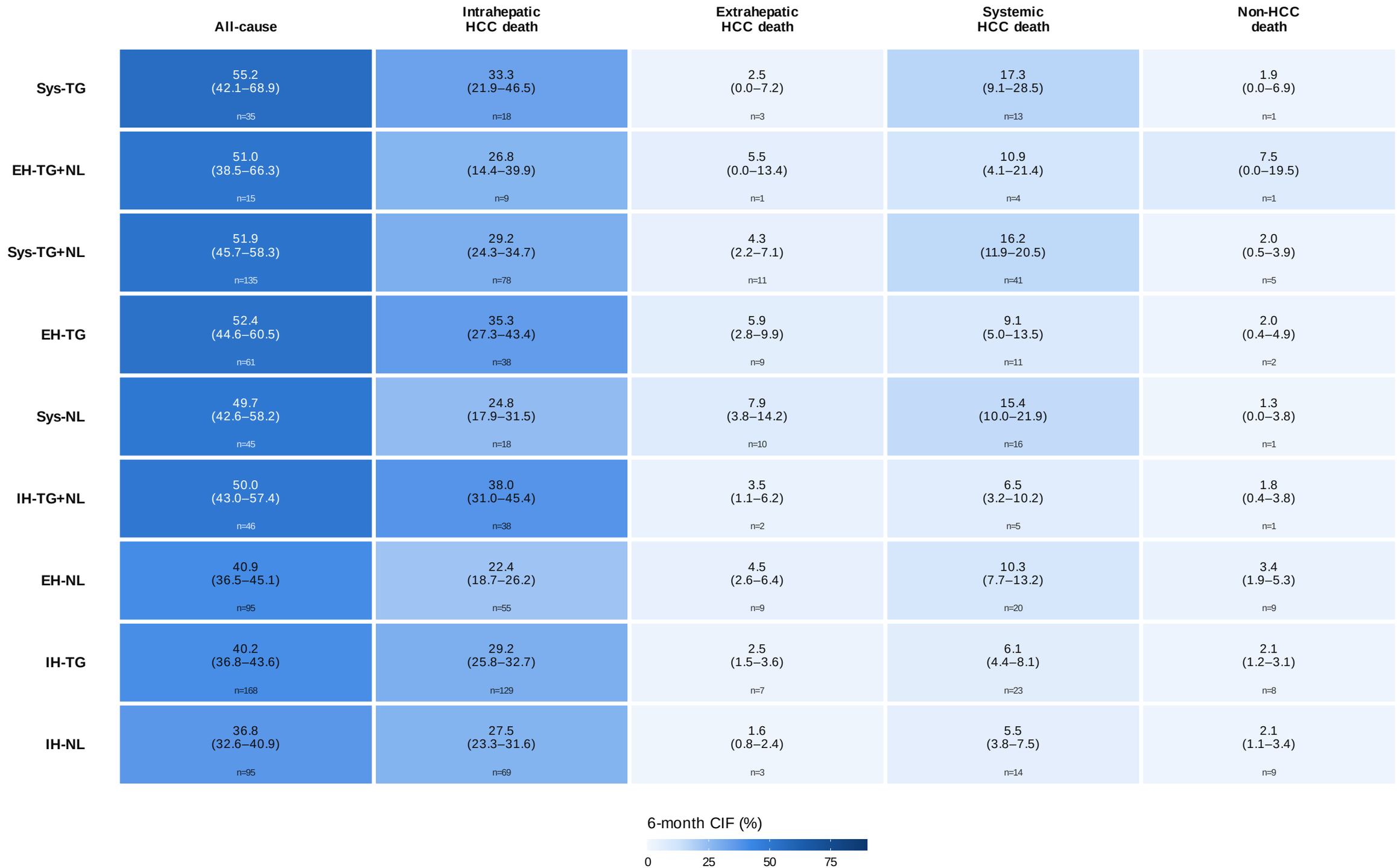
% (95% CI); all-cause = 1 − Kaplan–Meier; n = events within 12 months

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
Sys-TG	78.7 (62.3–88.0) n=36	39.0 (27.1–56.1) n=18	6.3 (2.1–18.9) n=3	31.1 (20.1–48.1) n=14	2.4 (0.3–16.3) n=1
EH-TG+NL	74.0 (51.2–86.2) n=21	35.0 (21.2–57.8) n=10	7.2 (1.9–27.3) n=2	20.9 (10.3–42.8) n=6	10.9 (3.7–31.7) n=3
Sys-TG+NL	79.5 (72.7–84.6) n=154	45.4 (38.9–53.1) n=88	6.8 (4.0–11.4) n=13	24.2 (18.8–31.1) n=47	3.1 (1.4–6.9) n=6
EH-TG	63.4 (53.1–71.4) n=72	36.4 (28.6–46.4) n=42	10.0 (5.7–17.5) n=11	13.5 (8.4–21.6) n=15	2.7 (0.9–8.1) n=3
Sys-NL	81.0 (69.2–88.2) n=63	33.0 (24.1–45.4) n=26	16.8 (10.2–27.7) n=13	28.5 (19.9–40.6) n=22	2.7 (0.7–10.6) n=2
IH-TG+NL	73.0 (62.3–80.7) n=74	53.8 (44.7–64.6) n=55	6.2 (2.9–13.6) n=6	9.9 (5.5–17.8) n=10	3.2 (1.0–9.7) n=3
EH-NL	62.9 (56.2–68.6) n=159	29.3 (24.2–35.5) n=76	7.3 (4.7–11.4) n=18	19.7 (15.3–25.5) n=48	5.9 (3.6–9.6) n=15
IH-TG	66.8 (61.7–71.3) n=269	47.4 (42.7–52.6) n=193	5.0 (3.2–7.7) n=19	9.6 (7.1–13.1) n=38	4.0 (2.5–6.5) n=16
IH-NL	49.8 (43.9–55.1) n=161	33.4 (28.6–39.0) n=109	2.6 (1.3–5.1) n=8	9.2 (6.5–13.0) n=29	4.7 (2.8–7.7) n=15



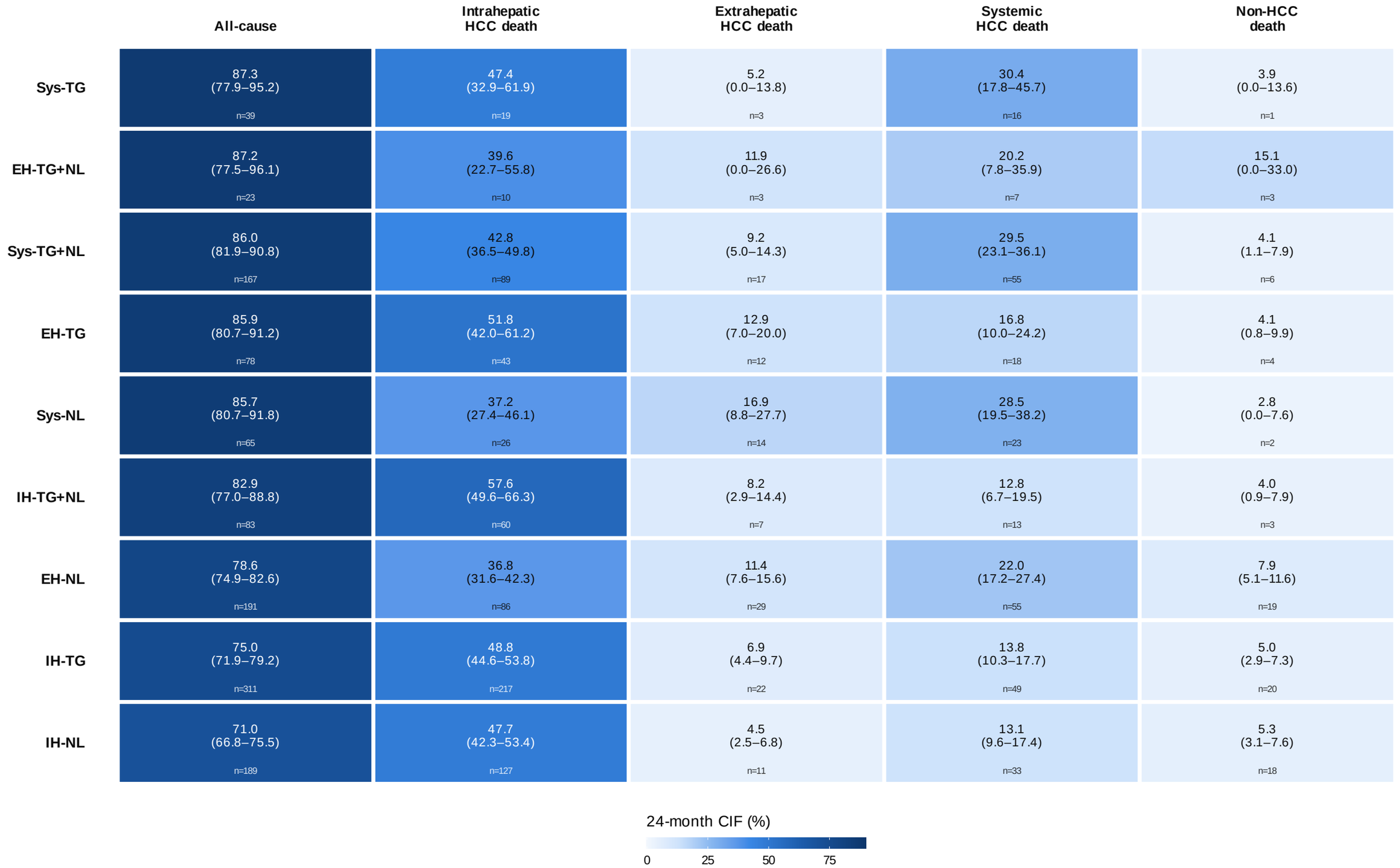
Supplementary: adjusted 6-month CIF after first progression

Same models as Figure 4.4A; n = events within 6 months



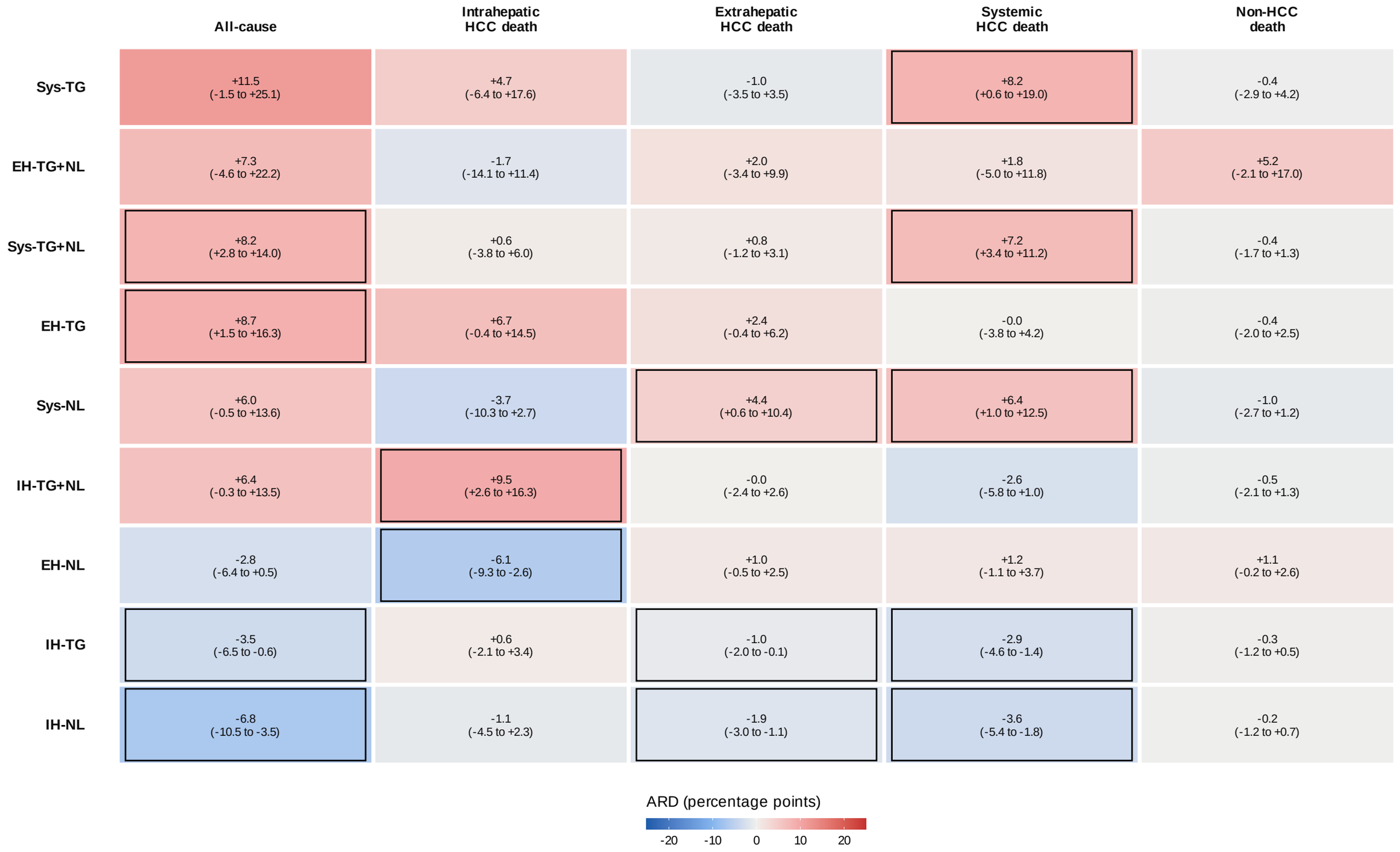
Supplementary: adjusted 24-month CIF after first progression

Same models as Figure 4.4A; n = events within 24 months



A2 Absolute risk difference at 6 months vs population-standardised average

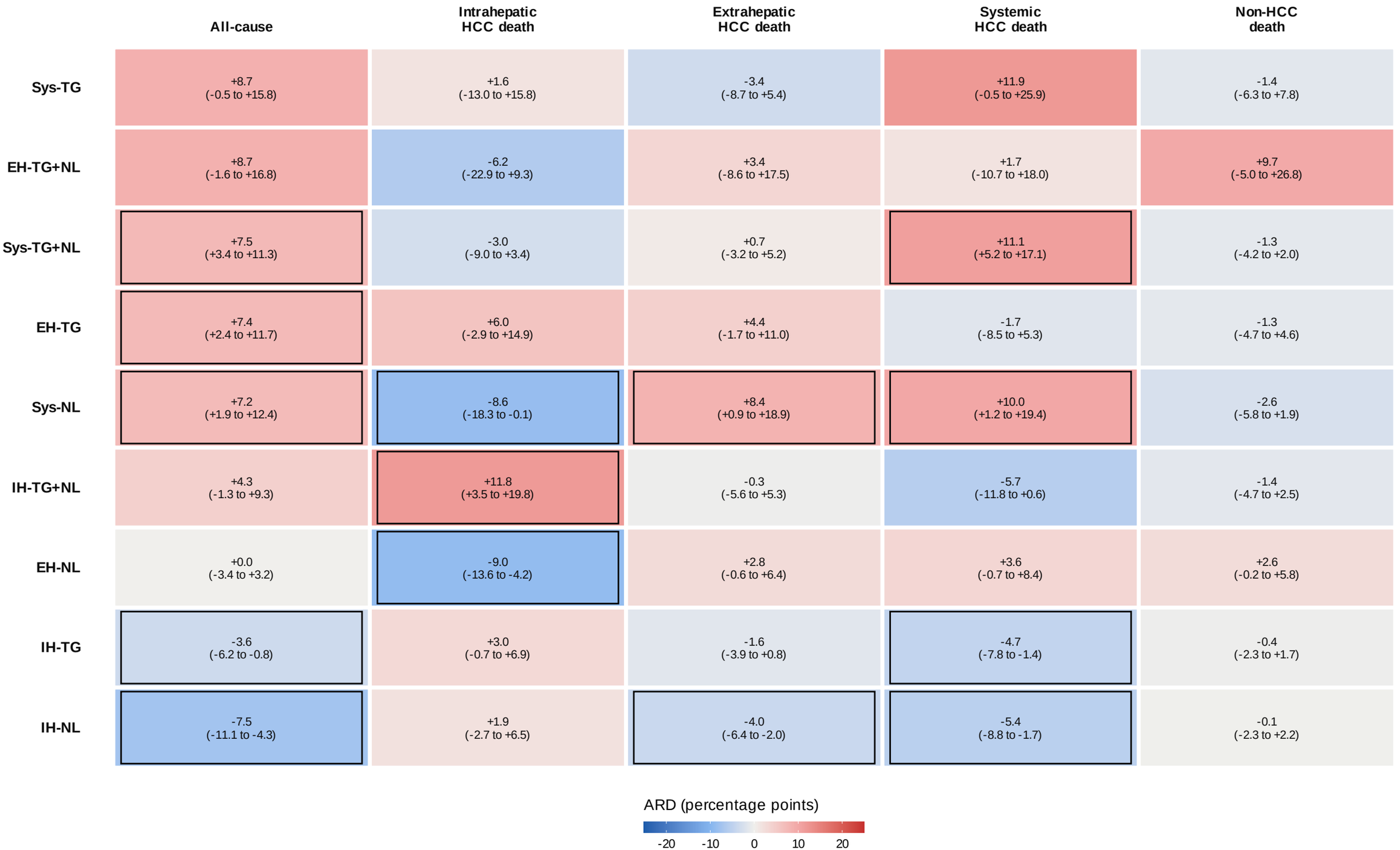
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

A2 Absolute risk difference at 24 months vs population-standardised average

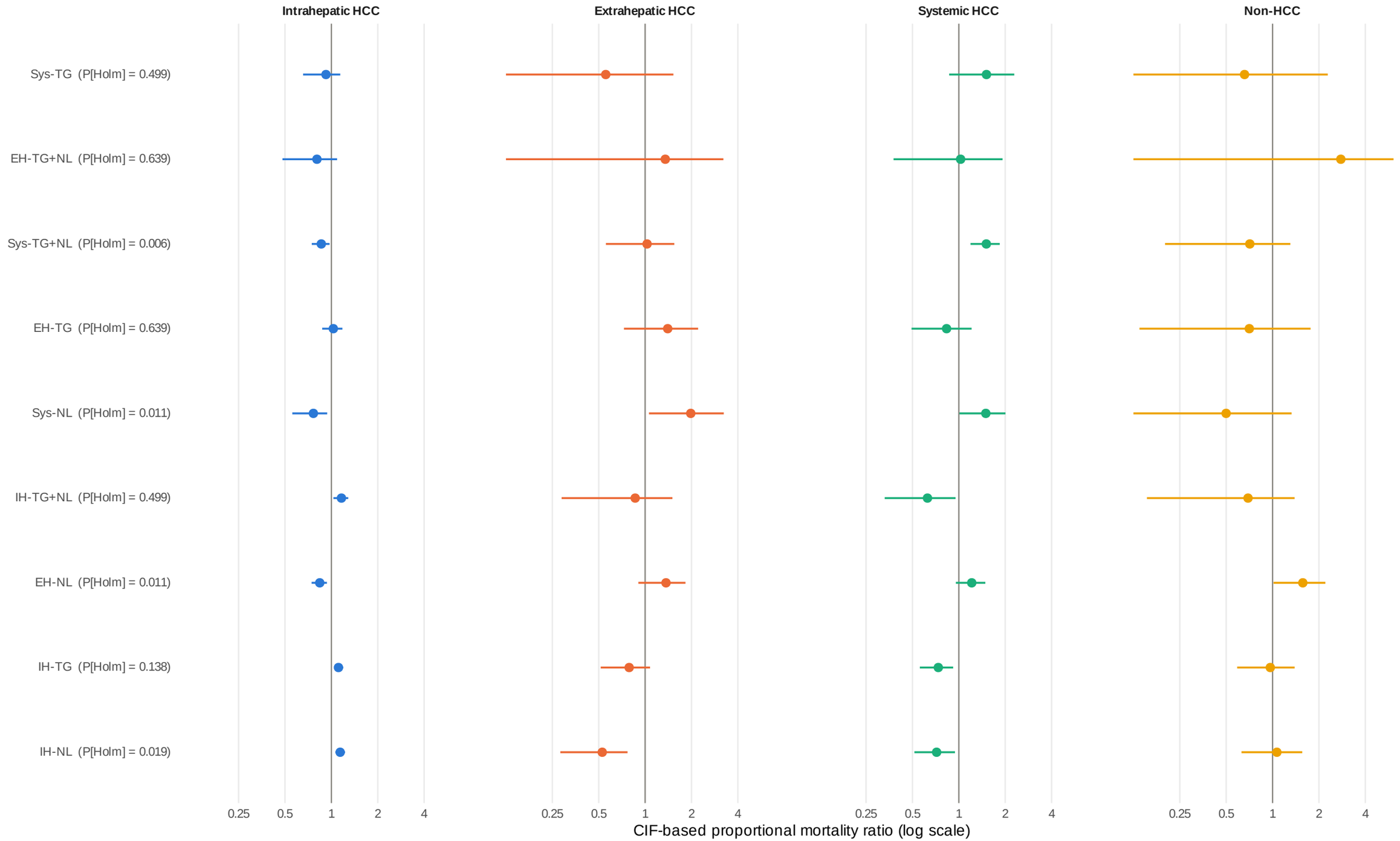
ARD = standardised CIF of the pattern − weighted average over patterns (weights = observed pattern shares); boxed: 95% CI excludes 0



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

A4 Cause-of-death enrichment: CIF-based proportional mortality ratio (PMR) at 6 months

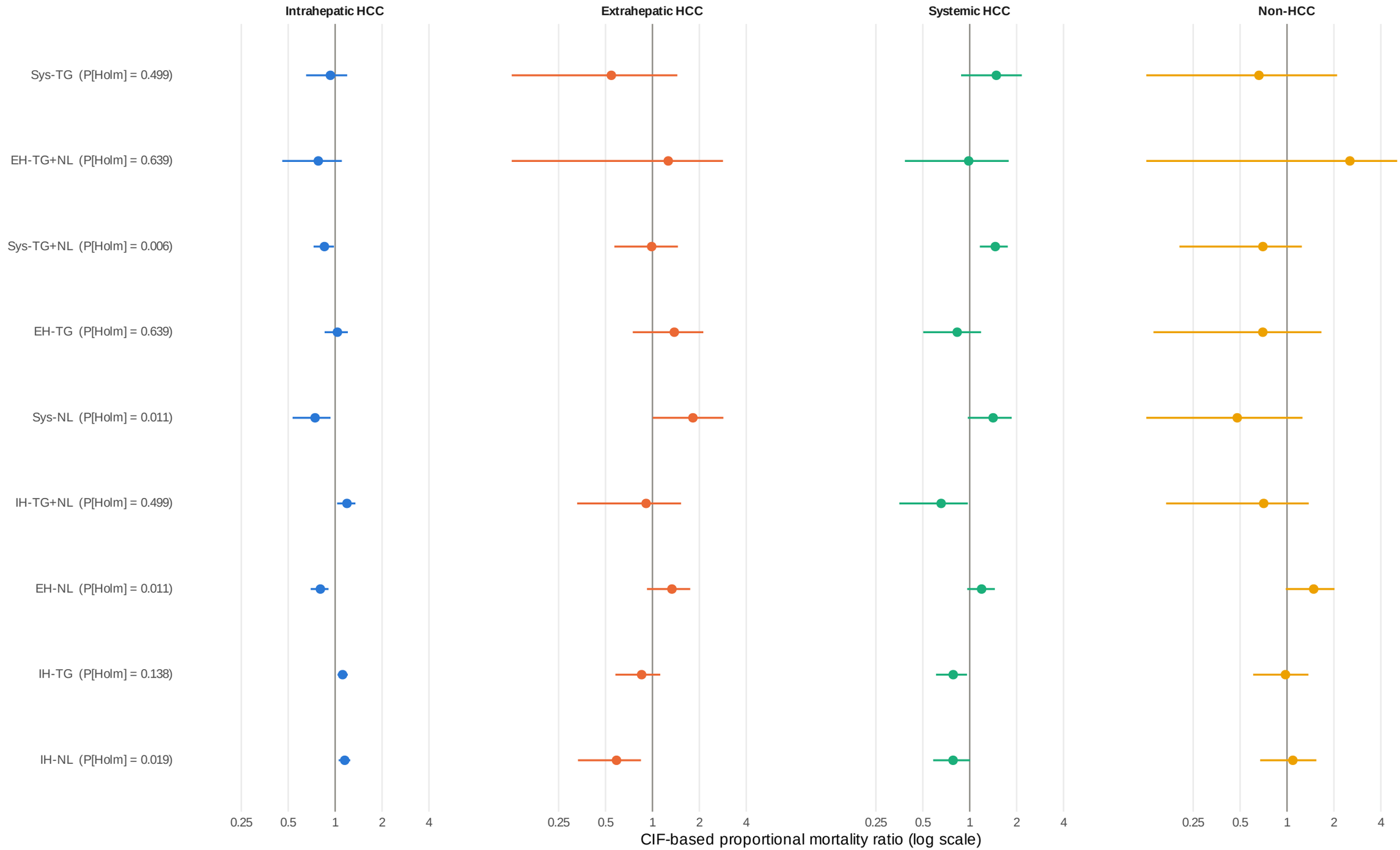
PMR = share of deaths from cause j within the pattern / share in all progressors. 95% bootstrap CI (clipped at 1/8 and 8).
P[Holm]: weighted-effect-coded Wald test (df = 2) that the pattern's csHR is equal across IH/EH/Systemic HCC death (from Lunn–McNeil model).



PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

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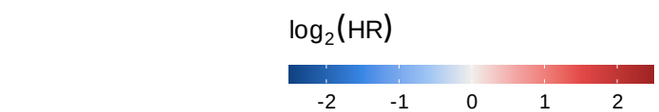


PMR is a proportion metric, not a risk metric: interpret alongside the absolute CIF matrix; PMRs within a pattern are mutually constrained.

Supplementary: Firth-penalised HR / csHR (reference: Sys-TG)

coxphf (Firth-penalised partial likelihood, Wald CI from penalised information); treatment group entered as covariate (coxphf has no strata)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.57 (0.40–0.80)**	0.68 (0.43–1.10)	0.76 (0.23–2.49)	0.29 (0.16–0.52)***	NE (sparse reference)
IH-NL	0.49 (0.34–0.69)***	0.62 (0.38–1.01)	0.45 (0.13–1.53)	0.25 (0.13–0.46)***	NE (sparse reference)
IH-TG+NL	0.75 (0.51–1.10)	0.98 (0.59–1.65)	1.16 (0.31–4.39)	0.36 (0.17–0.76)**	NE (sparse reference)
EH-TG	0.88 (0.60–1.29)	0.96 (0.56–1.65)	2.01 (0.59–6.84)	0.51 (0.26–1.01)	NE (sparse reference)
EH-NL	0.61 (0.43–0.87)**	0.52 (0.32–0.86)*	1.37 (0.43–4.42)	0.48 (0.27–0.86)*	NE (sparse reference)
EH-TG+NL	0.86 (0.51–1.45)	0.76 (0.36–1.64)	2.18 (0.45–10.56)	0.59 (0.24–1.46)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.85 (0.57–1.27)	0.67 (0.37–1.20)	2.78 (0.80–9.64)	0.83 (0.43–1.59)	NE (sparse reference)
Sys-TG+NL	0.87 (0.61–1.24)	0.81 (0.49–1.33)	1.41 (0.42–4.68)	0.88 (0.50–1.56)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.002**	**P < 0.001**	**P < 0.001**	**P = 0.303**

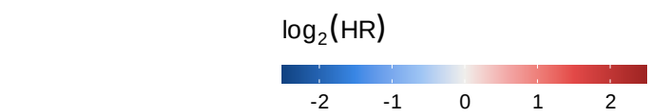


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR excluding undetermined deaths (reference: Sys-TG)

n = 1778 (8 deaths with unclassified/unknown/missing cause removed)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.58 (0.41–0.81)**	0.72 (0.45–1.17)	0.89 (0.26–3.08)	0.29 (0.16–0.52)***	NE (sparse reference)
IH-NL	0.50 (0.35–0.71)***	0.65 (0.40–1.07)	0.53 (0.15–1.91)	0.25 (0.13–0.46)***	NE (sparse reference)
IH-TG+NL	0.78 (0.53–1.15)	1.07 (0.64–1.81)	1.33 (0.33–5.31)	0.34 (0.16–0.73)**	NE (sparse reference)
EH-TG	0.89 (0.61–1.32)	1.02 (0.59–1.76)	2.35 (0.66–8.39)	0.49 (0.25–0.98)*	NE (sparse reference)
EH-NL	0.62 (0.44–0.88)**	0.54 (0.33–0.90)*	1.58 (0.47–5.35)	0.48 (0.27–0.85)*	NE (sparse reference)
EH-TG+NL	0.85 (0.50–1.43)	0.73 (0.34–1.60)	2.15 (0.41–11.34)	0.56 (0.22–1.43)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.84 (0.56–1.26)	0.67 (0.37–1.21)	3.08 (0.84–11.22)	0.81 (0.42–1.55)	NE (sparse reference)
Sys-TG+NL	0.88 (0.62–1.26)	0.82 (0.50–1.36)	1.69 (0.48–5.92)	0.89 (0.50–1.57)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.419**

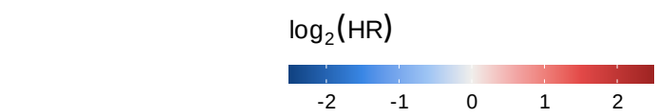


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR including patients flagged for deletion (reference: Sys-TG)

n = 1797

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.58 (0.41–0.81)**	0.70 (0.43–1.13)	0.88 (0.25–3.03)	0.28 (0.16–0.50)***	NE (sparse reference)
IH-NL	0.50 (0.35–0.72)***	0.65 (0.39–1.05)	0.52 (0.15–1.87)	0.24 (0.13–0.45)***	NE (sparse reference)
IH-TG+NL	0.79 (0.54–1.16)	1.07 (0.63–1.80)	1.33 (0.33–5.30)	0.35 (0.16–0.73)**	NE (sparse reference)
EH-TG	0.90 (0.61–1.33)	1.02 (0.59–1.76)	2.35 (0.66–8.40)	0.49 (0.25–0.99)*	NE (sparse reference)
EH-NL	0.63 (0.44–0.89)**	0.54 (0.33–0.90)*	1.55 (0.46–5.26)	0.47 (0.27–0.84)*	NE (sparse reference)
EH-TG+NL	0.85 (0.50–1.43)	0.73 (0.34–1.60)	2.17 (0.41–11.41)	0.57 (0.23–1.45)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.84 (0.56–1.26)	0.66 (0.37–1.20)	3.07 (0.84–11.19)	0.80 (0.42–1.55)	NE (sparse reference)
Sys-TG+NL	0.87 (0.61–1.24)	0.80 (0.48–1.32)	1.67 (0.48–5.83)	0.86 (0.48–1.52)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P = 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.392**

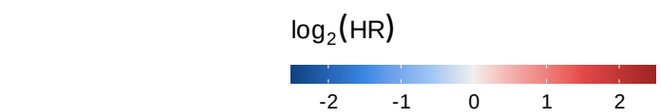


Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.

Supplementary: HR / csHR with the earlier pattern rule (reference: Sys-TG)

STG + TTU and EHG + TTU classified as Sys-TG instead of Sys-TG+NL (22 patients reclassified)

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.54 (0.40–0.72)***	0.63 (0.43–0.92)*	0.75 (0.25–2.24)	0.28 (0.16–0.47)***	NE (sparse reference)
IH-NL	0.46 (0.34–0.62)***	0.57 (0.38–0.84)**	0.45 (0.14–1.40)	0.24 (0.14–0.42)***	NE (sparse reference)
IH-TG+NL	0.73 (0.52–1.02)	0.94 (0.61–1.46)	1.14 (0.32–3.99)	0.34 (0.16–0.68)**	NE (sparse reference)
EH-TG	0.84 (0.60–1.19)	0.90 (0.56–1.43)	2.01 (0.64–6.26)	0.48 (0.25–0.92)*	NE (sparse reference)
EH-NL	0.58 (0.43–0.78)***	0.48 (0.31–0.72)***	1.35 (0.46–3.95)	0.47 (0.28–0.78)**	NE (sparse reference)
EH-TG+NL	0.79 (0.48–1.29)	0.65 (0.31–1.33)	1.84 (0.39–8.67)	0.56 (0.23–1.36)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.78 (0.55–1.12)	0.59 (0.35–0.98)*	2.63 (0.83–8.33)	0.79 (0.43–1.45)	NE (sparse reference)
Sys-TG+NL	0.79 (0.58–1.07)	0.68 (0.44–1.03)	1.43 (0.47–4.35)	0.85 (0.50–1.44)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.397**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group.